YASHU SHUKLA

Data Science · AI Research Engineer

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CONTACT

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ABOUT ME

I am an AI Research Engineer passionate about Physical AI and dexterous robotics. As a Data Science graduate from Hanyang University, my work sits at the frontier of Robot Foundation Models and Vision-Language-Action systems; spanning 3D scene-aware perception, tactile-torque sensory integration, and touch-aware dexterous manipulation policies. Beyond robotics, I bring strong hands-on experience in machine learning, NLP, and statistical modeling, having built and deployed ML systems across AI robotics labs and IT startups; from patent-filed anomaly detection to sentiment analysis at scale. I am driven to push the boundaries of what intelligent systems can do in the physical world.

EDUCATION

Bachelors of Engineering in Data Science | Hanyang University, South Korea **2022 - 2026**

Major: Data Science Engineering and Physiological Brain Science
GPA: 4.0 out of 4.5

Middle and High School | Vibgyor High School, India **2015 - 2022**

Major: Science (Mathematics, Physics, Chemistry, English, Physical Education)
- 2022- Ranked in the Top 5 in Grade 12 (92.6%)
- 2020- Ranked in the Top 10 in Grade 10 (93.5%)

Primary School | International School of Koje, South Korea **2008 - 2015**

- Studied in an international environment
- Experience of solving practical case studies

TECHNICAL SKILLS

Languages: Python, C, C++, R, JavaScript, Typescript
Frameworks / Tools: TensorFlow, PyTorch, scikit learn, Keras, NumPy, pandas, Matplotlib, Seaborn, Django
Robotics & Simulation: Isaac Sim/Lab, Robot Manipulation, 3D Object Scanning, ROS
Development & Analytics Tools: Angular 10, ELK Stack, PsychoPy, API development
Databases: SQL, PostgreSQL
Cloud & Deployment: AWS, GCP
Productivity Tools: MS Office

EXPERIENCE

1) AI Research Engineer Intern - RLWRLD, Seoul **Dec 2025-Present**

- Built and curated robotic manipulation datasets in Isaac Sim, optimized dataset composition to achieve equivalent model accuracy with 50% reduced data volume.
- Fine-tuned and systematically benchmarked multiple VLA models, establishing performance baselines for robot foundation model training.
- Developed end-to-end automated training and evaluation pipeline reducing model setup time from 20–30 minutes to under 30 seconds.
- Integrating 3D geometry perception into VLA models via foundation model-based scene understanding, enabling multimodal sensor fusion (tactile, joint torque) for touch-aware dexterous manipulation policies.

2) Data Scientist - w3company, Seoul **March 2025-Dec 2025**

(Part-time)

- Part of the R&D team developing machine learning–driven multi-module system for fake account detection and anomaly detection; (patent filed)
- Built and evaluated deep learning image classifiers using Residual Networks (ResNet) for synthetic image detection, targeting AI-generated profile photos and fraudulent user identities
- Developed and evaluated BERT-based NLP models for sentiment analysis and text classification to support improvements in a social networking app (Moji) and provide user engagement insights

3) Undergraduate Thesis Research **March 2025-Dec 2025**

(Advisor: Prof. Noh Youngtae, Department of Data Science)

- Research aimed to enable non-invasive continuous health monitoring by addressing noise interference in mmWave FMCW radar (AWR1642BOOST) vital sign signals.
- Developed a CNN+BiLSTM+Attention model for contactless heart rate and breathing rate estimation from mmWave radar signals; 3× improvement over classical signal processing baselines.

4) Teaching Assistant - Hanyang University, Seoul **Sep. 2024-Dec. 2024**

- Managed and tracked a class of over 30 students for the course “Understanding Business Environment
- Helped facilitate group discussions and ensured smooth execution of class activities

5) Software Engineer Intern - Bespoke System Pvt. Ltd., India **July 2024-Aug. 2024**

- Worked on web development using ASP.NET framework, MS SQL, Angular 10, Web API, and SQL Server
- Gained experience in both frontend and backend development, focusing on API integration and database management

PUBLICATION & CONFERENCES

1) Research Contributor to the "RLDX-1 Technical Report" (arXiv) **Seoul, May 2026**

*Dongyoung Kim,Myungkyu Koo, ...**Yashu Shukla**, ... Yeonjae Lee, Jinwoo Shin*

- Contributed to research on Robot Foundation Models and Vision-Language-Action (VLA) systems for dexterous humanoid robot manipulation
- Developed an end-to-end automated pipeline for VLA model training and inference

2) The 12th Swiss-Korean Life Science Symposium, Research Attendee **Seoul, May 2025**

3) Presented research poster, **"Cerebellum Activation in Depression"** at the Korean Society for Brain and Neural Sciences (**KSBNS**) Conference **Seoul, Aug. 2025**

PROJECTS

1) Stress Detection of Drivers Using mmWave Radar **March 2025 - Dec 2025**

- Developed a stress detection system using Texas Instruments' IWR1642BOOST mmWave sensor to monitor vital signs(HR and BR) behavior non-contactless
- Analyzed and denoised the collected vital sign data to extract meaningful vital related features

2) StressMate – Mental Wellness Web App **May 2025 - June2025**

- Built a Django-based app integrating LLMs (Perplexity AI, Gemini) along with RAG
- Deployed on AWS using EC2 and S3 for scalable hosting and cloud based storage

3) Cerebellar fMRI Analysis in Major Depressive Disorder **May 2025 - June2025**

- Conducted advanced fMRI analysis revealing significantly reduced cerebellar response to positive stimuli in clinical depression patients compared to neurotypical participants, highlighting emotion-processing deficits

5) PlateCheck Application **Oct. 2024 - Dec. 2024**

- Developed LLM based Korean Food Ingredient Recognition that accommodates both food image and text inputs using Django
- Integrated a fine-tuned ResNet model for image-based dish identification and utilized GPT-4o Mini for extracting relevant ingredient details.
- Conducted and evaluated the service based on HCI factors via user study

6) Food Demand Forecasting using Temporal Convolutional Networks **Oct. 2024 - Dec. 2024**

- Built a time series forecasting model using Temporal Convolutional Network (TCN) to forecast food demand based on historical sales data

7) News Audio Summarizer using BART **April 2024 - June 2024**

- Designed an NLP-based system for converting audio news content to text and generating concise summaries by fine tuning the BART model on transcribed news articles paired with human written summaries

9) Sign Language Recognition Model with MLP and CNN **Nov. 2023**

- Developed a machine learning model to recognize sign language gestures, leveraging both Multi-Layer Perceptrons (MLP) and Convolutional Neural Networks (CNN)

10) Clothing Size Prediction Model for Fashion E-Commerce **April 2022 - June 2022**

(Group Project)

- Developed Clothing Size Prediction Model for Fashion E-Commerce with Machine Learning Approaches
- Collected data from the MUSINSA online clothing store using Python Selenium

LANGUAGES

- **Native:** English and Hindi
- **Intermediate:** Gujarati
- **Beginner:** Korean

SCHOLARSHIPS AND ACHIEVEMENTS

- Excellence Award, College of Engineering, Hanyang University **August 2026**
- Hanyang Brain (Academic Excellence) Scholarship **March 2026 - June 2026**
- Daewoong Foundation Global Scholarship **Nov. 2025 - March 2026**
- Hanyang Brain (Academic Excellence) Scholarship **Sep. 2025 - Dec. 2024**
- Hanyang Brain (Academic Excellence) Scholarship **March 2025 - June 2025**
- Special Scholarship for Academic Encouragement Hanyang (Foreigner) **Sep. 2024 - Dec. 2024**

EXTRACURRICULAR

- Maintains a personal blog on achieving balanced work and lifestyle
- Completed a 5 year Senior Diploma in Kathak (Indian Classical Dance)
- Kathak Performer – Indian Embassy Cultural Events, South Korea
- Engaged in team sports including basketball and badminton